

# Similar Triangles in Combination

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*missing sides, scale factor, perimeter ratio and area ratio*

## Directions

- If two triangles are similar with scale factor  $k$  (target length over starting length), then every side is multiplied by  $k$ , the perimeter is multiplied by  $k$ , and the area is multiplied by  $k^2$ .
- Decide which of those three relationships a problem needs before you compute — the sheet mixes them deliberately.
- Give lengths and perimeters to one decimal place where needed, and include the unit on every measurement answer.

**1.**  $\triangle ABC \sim \triangle DEF$  with  $AB = 5$  cm,  $BC = 7$  cm and  $DE = 15$  cm. Find the length of  $\overline{EF}$ , in centimetres.

**Answer:** \_\_\_\_\_ cm

**2.**  $\triangle PQR \sim \triangle STU$  with  $PQ = 12$  in,  $PR = 21$  in and  $ST = 8$  in. Find the length of  $\overline{SU}$ , in inches.

**Answer:** \_\_\_\_\_ in

**3.** Two similar triangular road signs are cut from the same pattern. The smaller sign has a base of 6 cm and the larger sign has a base of 15 cm. Find the scale factor from the smaller sign to the larger sign.

Answer: \_\_\_\_\_

**4.**  $\triangle ABC \sim \triangle DEF$  with  $AB = 8$  m and  $DE = 12$  m. The perimeter of  $\triangle ABC$  is 30 m. Find the perimeter of  $\triangle DEF$ , in metres.

Answer: \_\_\_\_\_ m

**5.** Two similar triangular floor tiles have corresponding sides of 6 cm and 10 cm. The smaller tile has an area of  $27 \text{ cm}^2$ . Find the area of the larger tile, in square centimetres.

Answer: \_\_\_\_\_  $\text{cm}^2$

**6.**  $\triangle ABC \sim \triangle DEF$ . The perimeter of  $\triangle ABC$  is 24 cm and the perimeter of  $\triangle DEF$  is 60 cm. If  $AB = 7$  cm, find the length of  $\overline{DE}$ , in centimetres.

**Answer:** \_\_\_\_\_ cm

**7.** Two similar triangular garden plots have areas of  $48 \text{ m}^2$  and  $108 \text{ m}^2$ . Find the ratio of a side of the larger plot to the matching side of the smaller plot.

**Answer:** \_\_\_\_\_

**8.** Two similar triangular sails have areas of  $50 \text{ cm}^2$  and  $128 \text{ cm}^2$ . The smaller sail has a perimeter of 30 cm.

(a) Find the perimeter of the larger sail, in centimetres.

**Answer:** \_\_\_\_\_ cm

(b) Justify why the perimeter ratio is the square root of the area ratio rather than the area ratio itself.

**9.** A triangular window is enlarged so that its area becomes 6.25 times the area of the original.

(a) Find the scale factor applied to each side.

Answer: \_\_\_\_\_

(b) The original window has a perimeter of 84 cm. Find the perimeter of the enlarged window, in centimetres.

Answer: \_\_\_\_\_ cm

**10.**  $\triangle ABC \sim \triangle DEF$  with  $AB = 14$  cm and  $DE = 21$  cm. The area of  $\triangle ABC$  is 56  $\text{cm}^2$ .

(a) Find the area of  $\triangle DEF$ , in square centimetres.

Answer: \_\_\_\_\_  $\text{cm}^2$

(b) Explain why multiplying every side of a triangle by the same factor multiplies its area by the square of that factor.